

# UNDERSTANDING GRAMMAR REPRESENTATION IN THE HUMAN BRAIN VIA INFORMATION-RESTRICTED LLMS

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# BACKGROUND

How does the brain distinguish nouns and verbs?



How does the brain distinguish nouns and verbs? **It's complicated...**



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## Fronto-Temporal Dichotomy

Damasio and Tranel (1993)

Patients with **brain lesions** doing **object** and **action** naming tasks

→ **Temporal** regions involved in the retrieval of **nouns**

→ **Frontal** regions involved in the retrieval of **verbs**

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*What is driving the distinction?*

How does the brain distinguish nouns and verbs? **It's complicated...**

## **Emergentist View**

Bird et al. (2000), Vigliocco et al. (2011)

Noun / Verb distinction is not **lexical** but **semantic**

→ Nouns are **more imageable** and related to **objects**

→ Verbs are **less imageable** and related to **actions**



How does the brain distinguish nouns and verbs? **It's complicated...**

## Combinatorial View

Shapiro and Caramazza (2003), Tyler et al. (2003)

Noun / Verb distinction driven by **computational** processes

→ Pseudowords: *he wugged* and *the wugs* represented in distinct brain areas

→ Verbs are more **computationally demanding**

How does the brain distinguish nouns and verbs? **It's complicated...**

## **Distributed Account**

Crepaldi et al. (2013)

Theory-blind meta-analysis based on 36 neuroimaging studies

→ 32 reliable **clusters**:

3 for **nouns** (R-ITG, L-AG, L-IPG)

1 for **verbs** (R-MTG)

3 both **nouns** and **verbs**; Task dependent (L-IFG, R-IFG, L-Insula)

→ No support for **Fronto-Temporal** dichotomy or **Embodied Cognition**

→ **Distributed** processing of nouns and verbs

# MOTIVATION



## Prior Methods

Picture Naming (Shapiro and Caramazza, 2011)

Semantic Categorization (Tyler et al., 2003)

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## Current Approach

Naturalistic Stimuli



## Stimuli

- 11 stories from  
*The Moth Radio Hour*
- 10-15 min per story
- Listening and reading

## fMRI Data

- From Deniz et al. (2019)
- 9 subjects

## Features

### Listening

- Phonemes, phoneme count

### Reading

- Letters, letter count
- Word length variation

### Both

- Number of words
- Number of pauses
- Lexical (English1000)
- Contextual Embeddings

# CONTEXT MASKING

## Stimulus

*There're some scientists who say that the universe that we inhabit is not the only universe there is ...*



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↓ POS tagging via spaCy

## Attention Mask

|          |      |            |     |     |      |      |          |       |    |
|----------|------|------------|-----|-----|------|------|----------|-------|----|
| There're | some | scientists | who | say | that | the  | universe |       |    |
| that     | we   | inhabit    | is  | not | the  | only | universe | there | is |

# CONTEXT MASKING

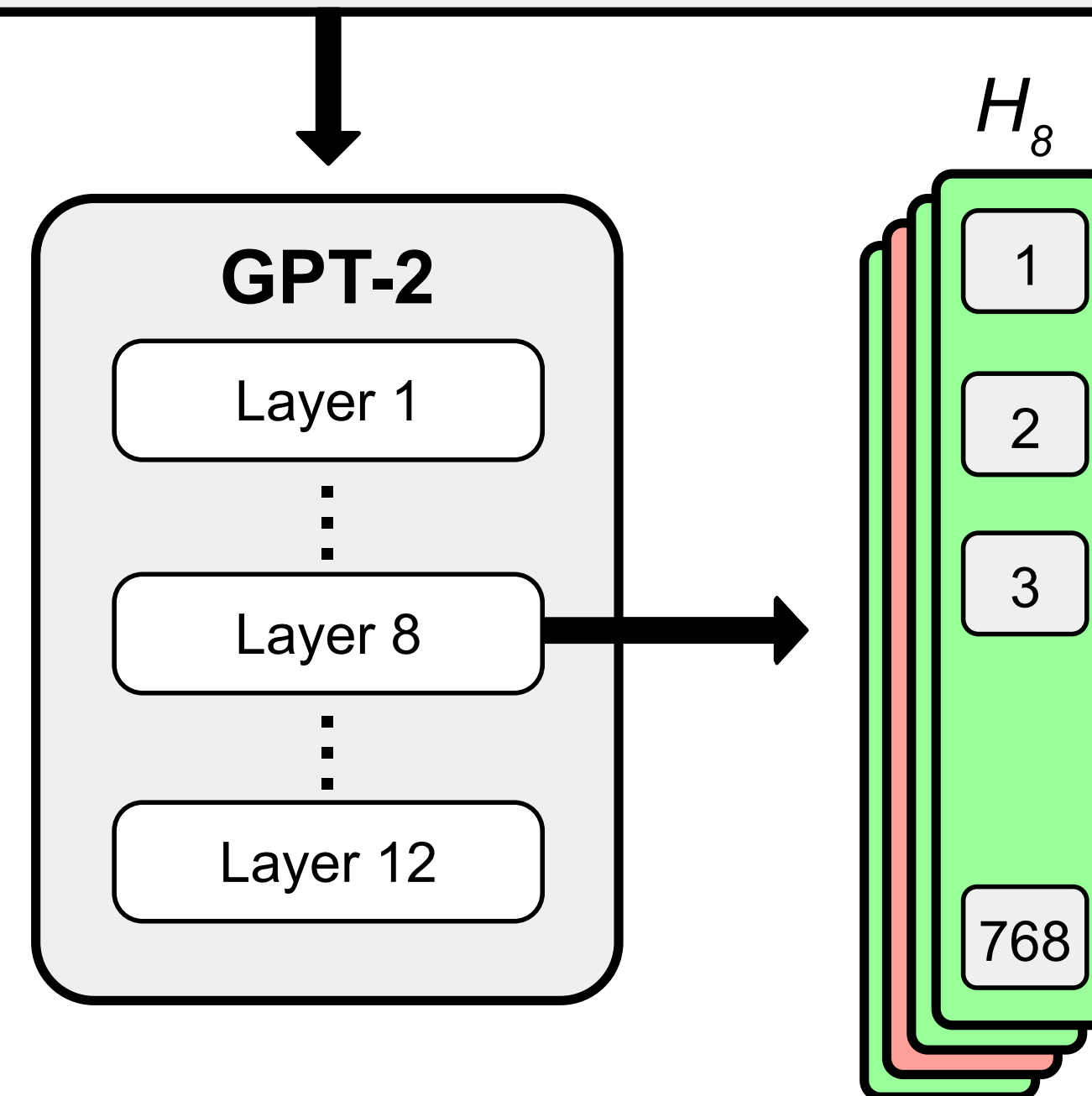
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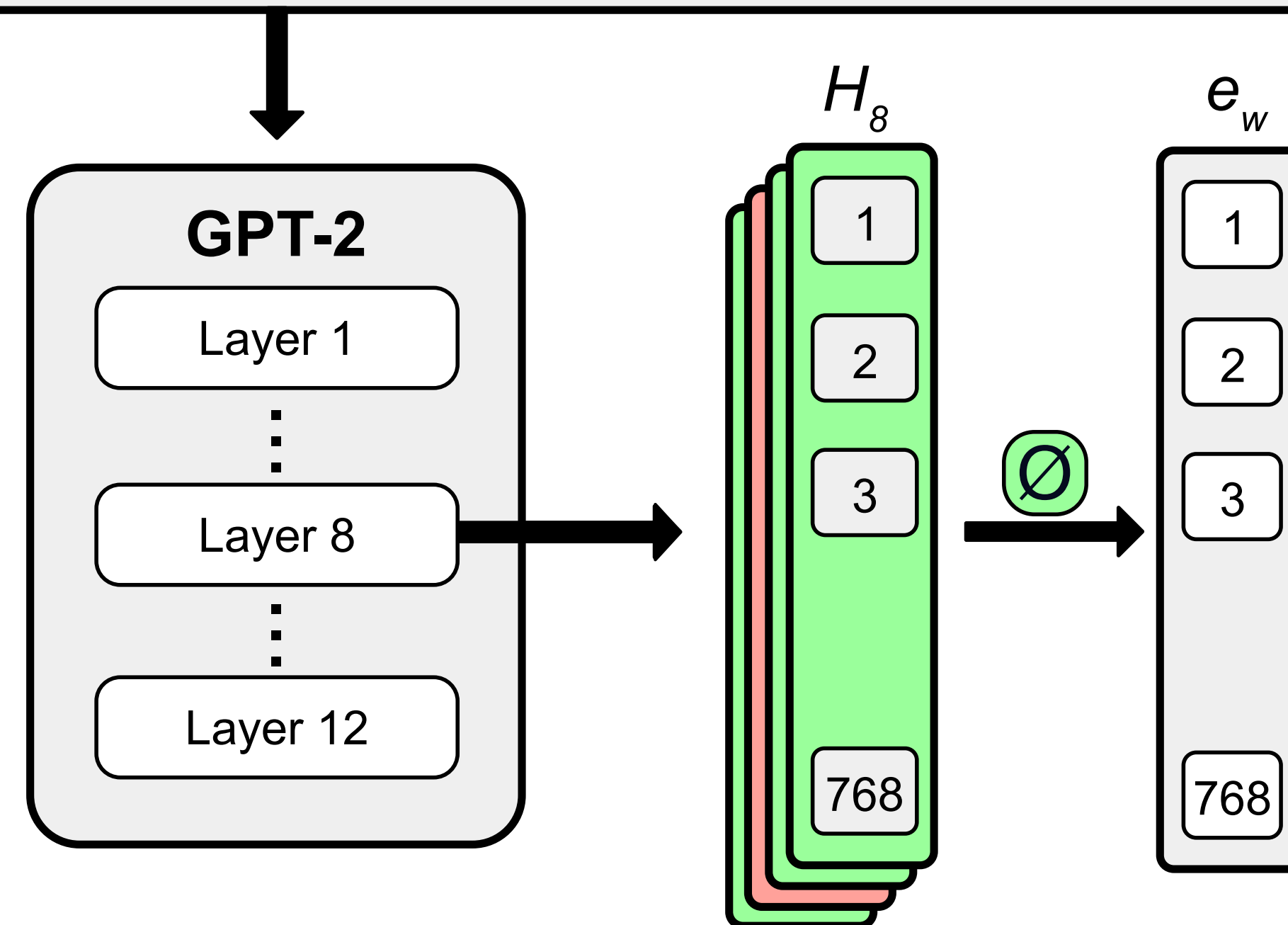
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# VOXEL-WISE ENCODING

## **Banded Ridge Regression**

to model BOLD as linear combination of masked embeddings and low-level features



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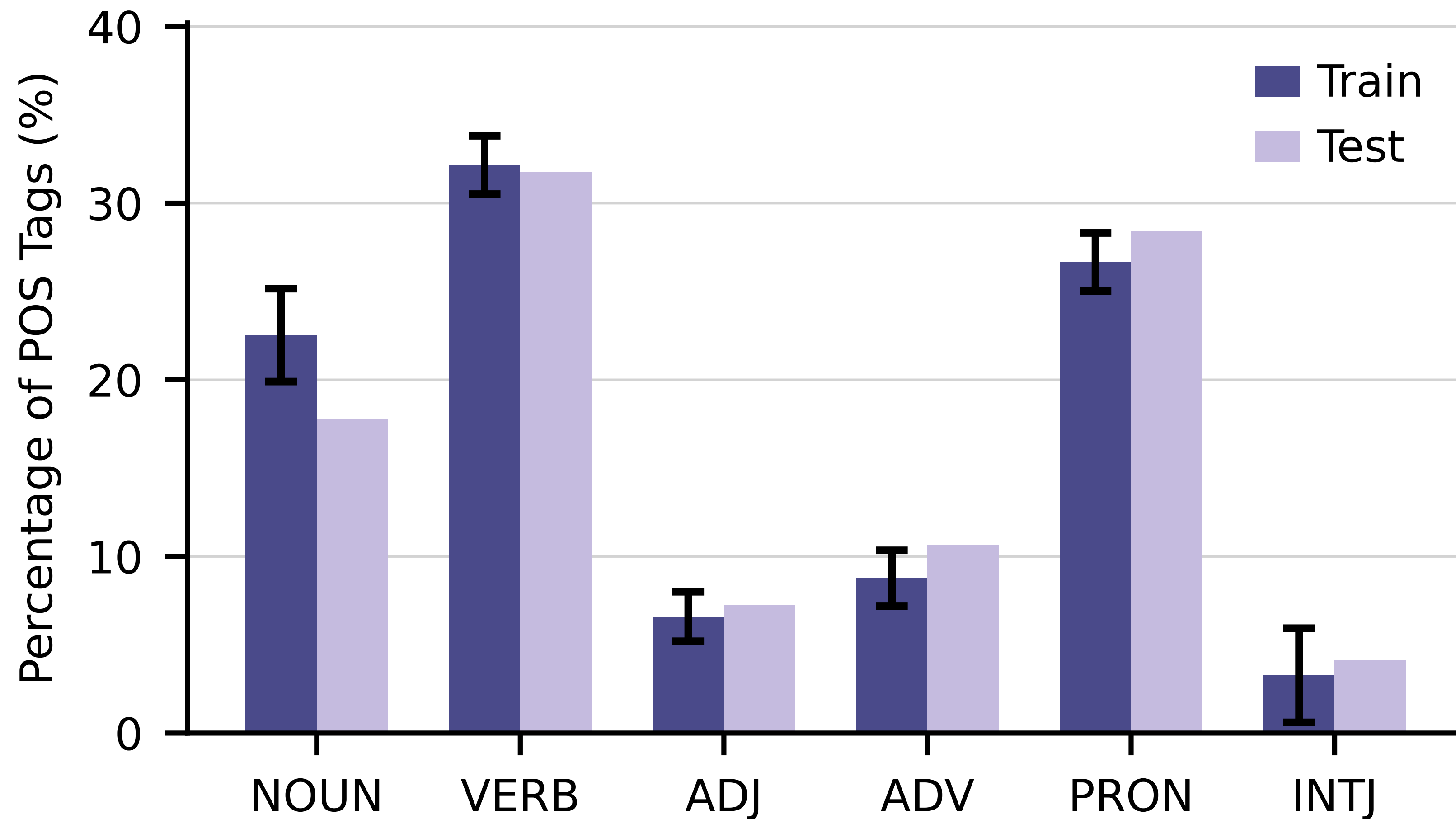
## **POS used**

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## **Prediction accuracy**

Pearson correlation between predicted and actual train data BOLD response

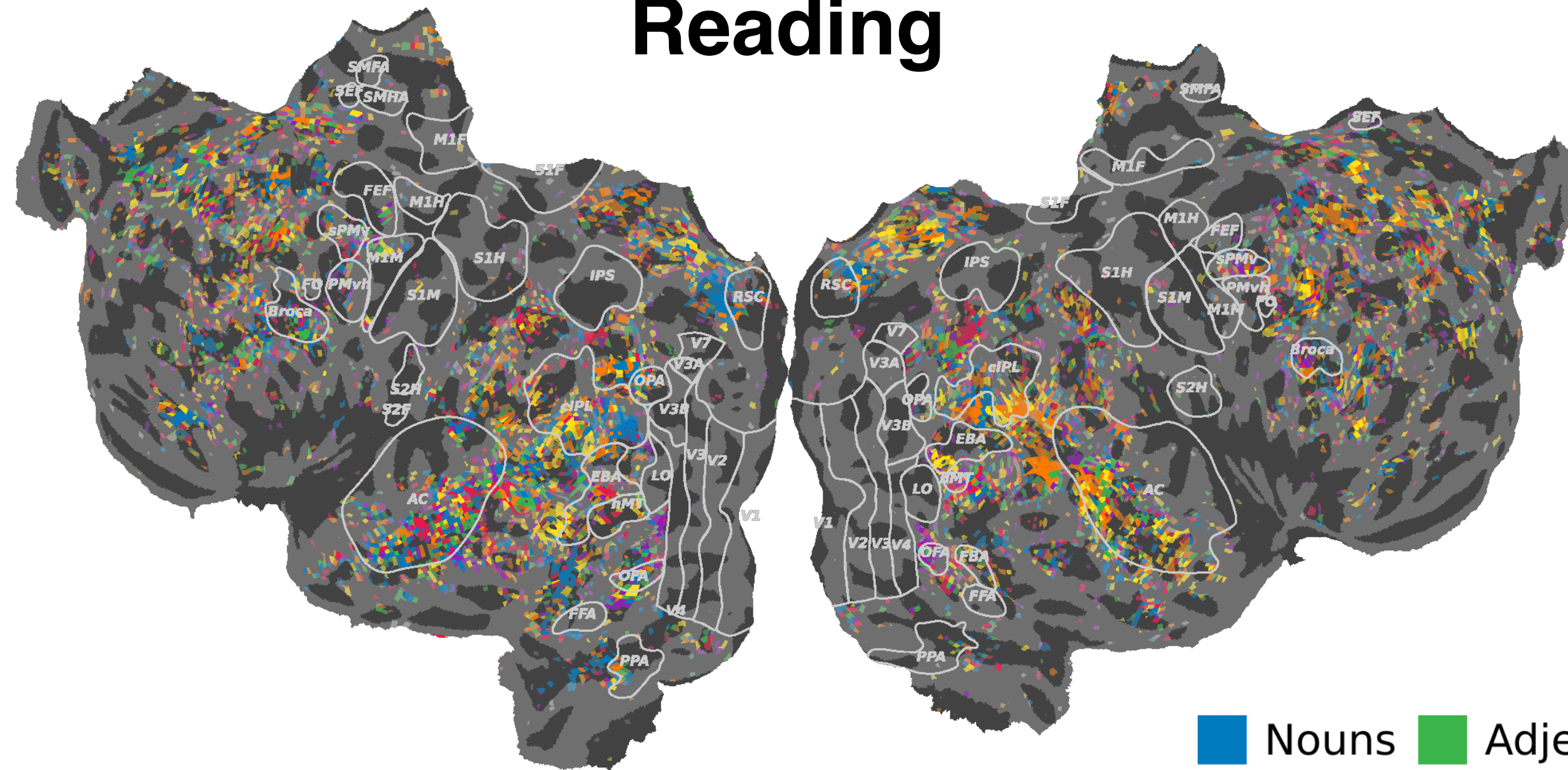
# PARTS OF SPEECH FREQUENCY IS CONSISTENT ACROSS STORIES



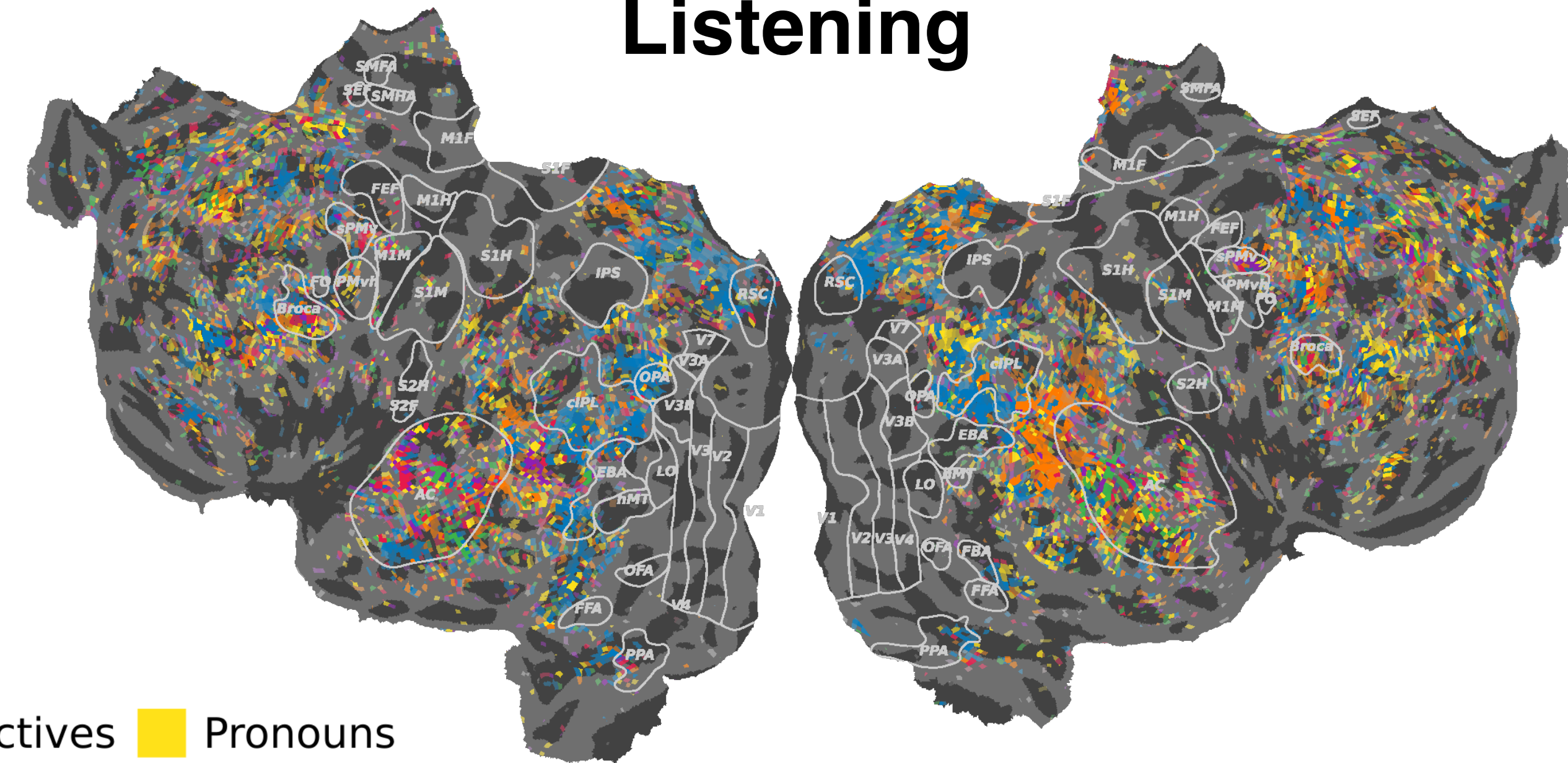


# POS-MASKED CONTEXTUAL EMBEDDINGS REVEAL CLUSTERS OF POS SENSITIVITY ACROSS THE BRAIN

Subject 7  
Reading



Subject 7  
Listening

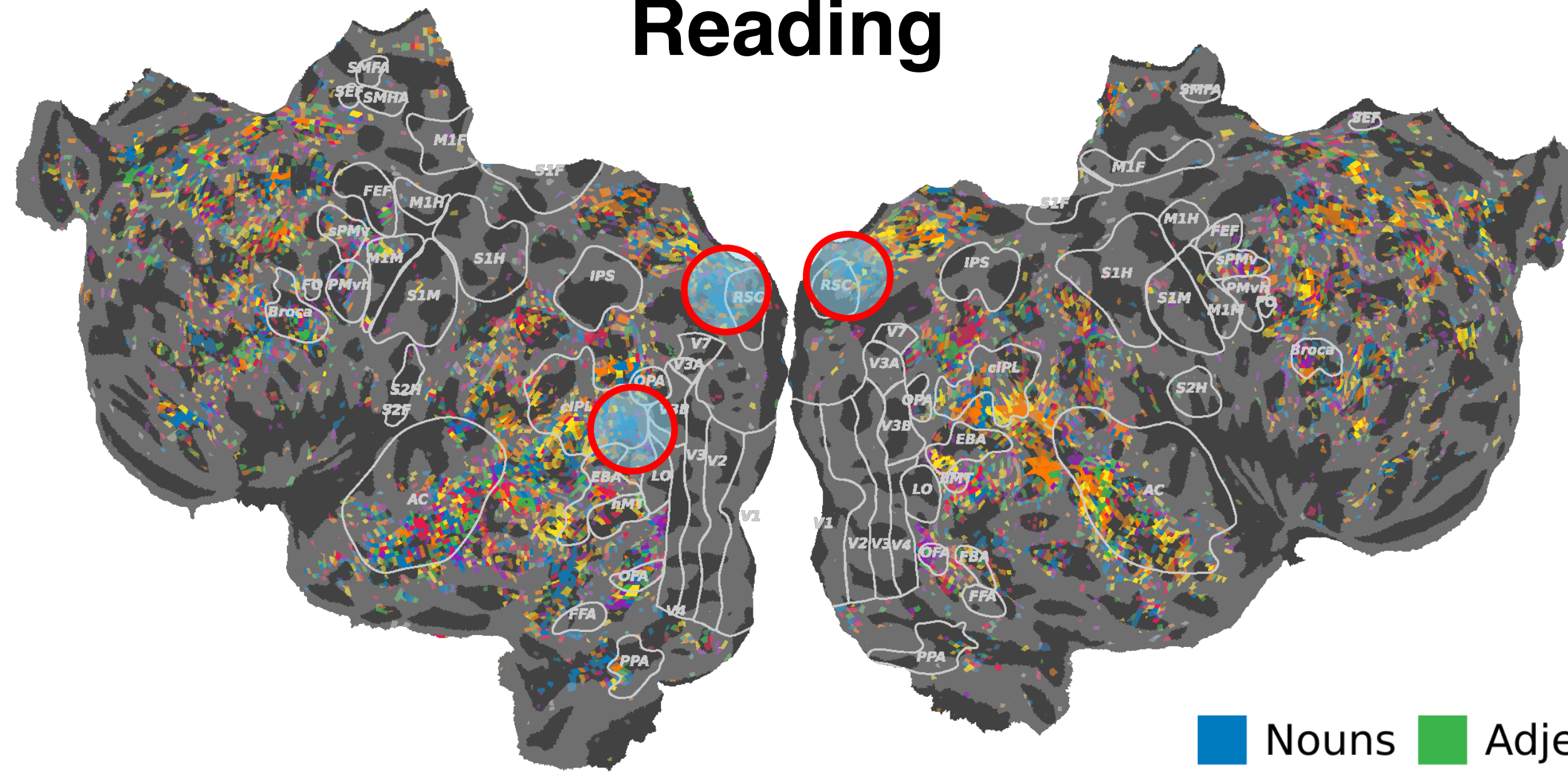


■ Nouns ■ Adjectives ■ Pronouns  
■ Verbs ■ Adverbs ■ Interjections

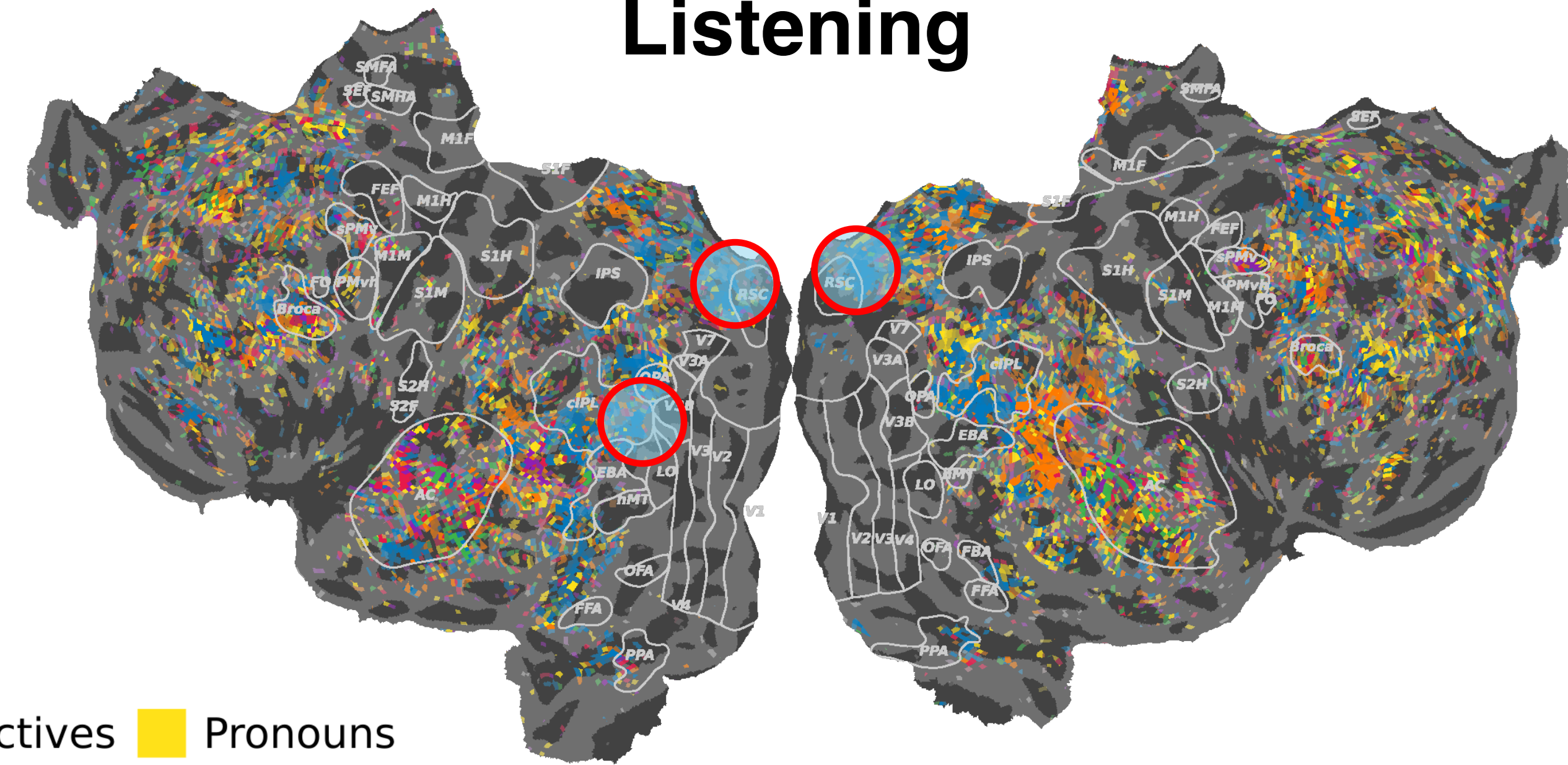


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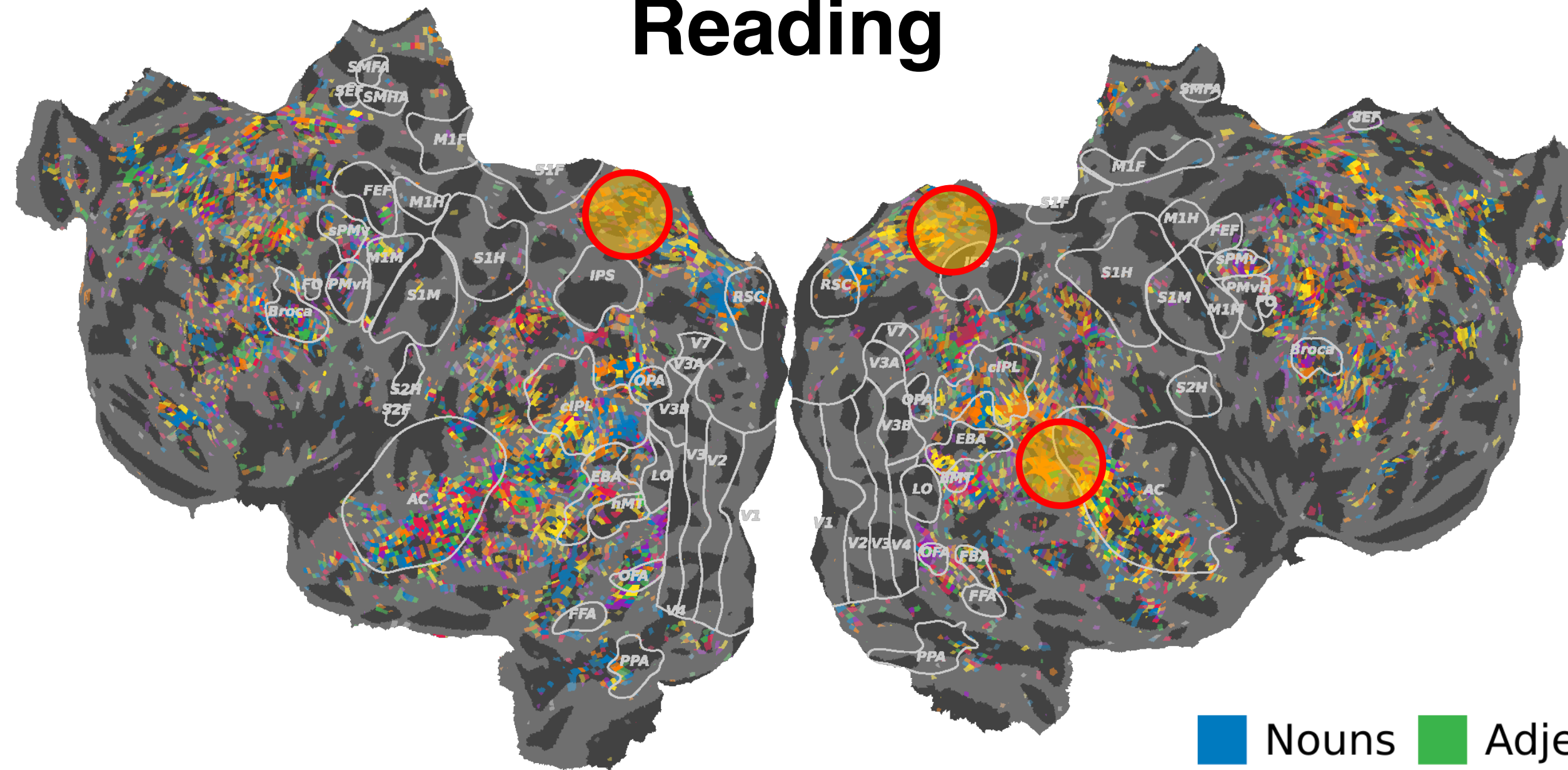


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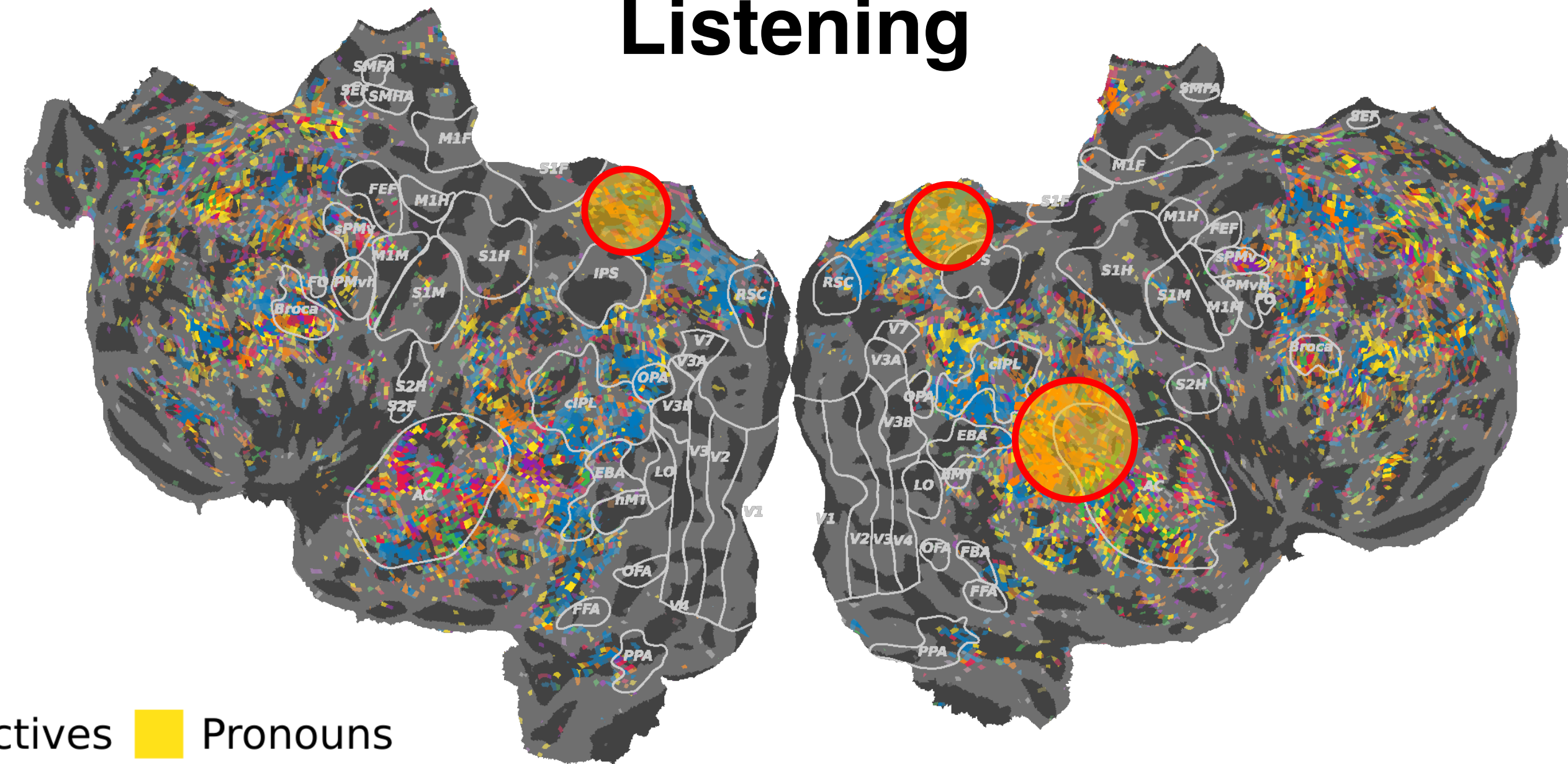


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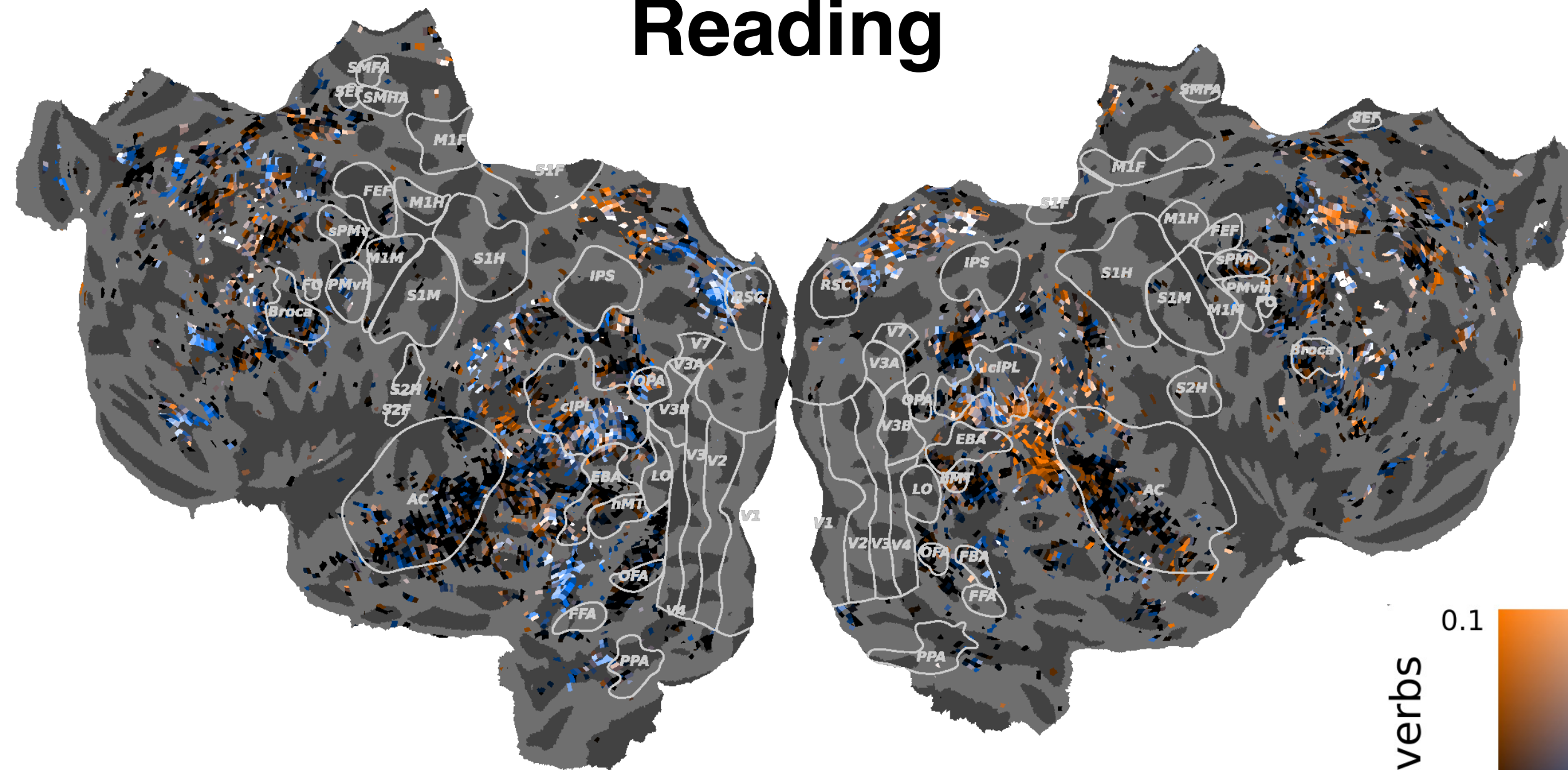


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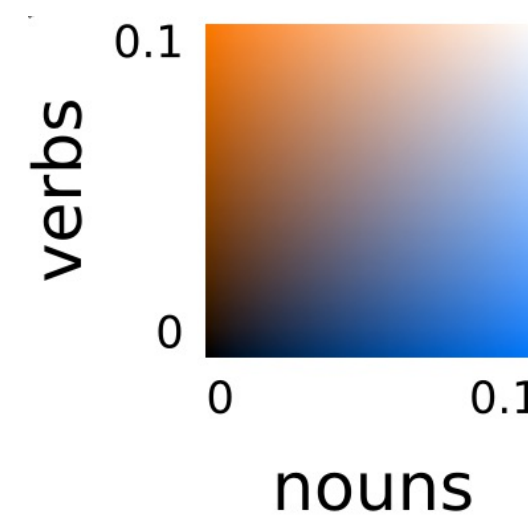
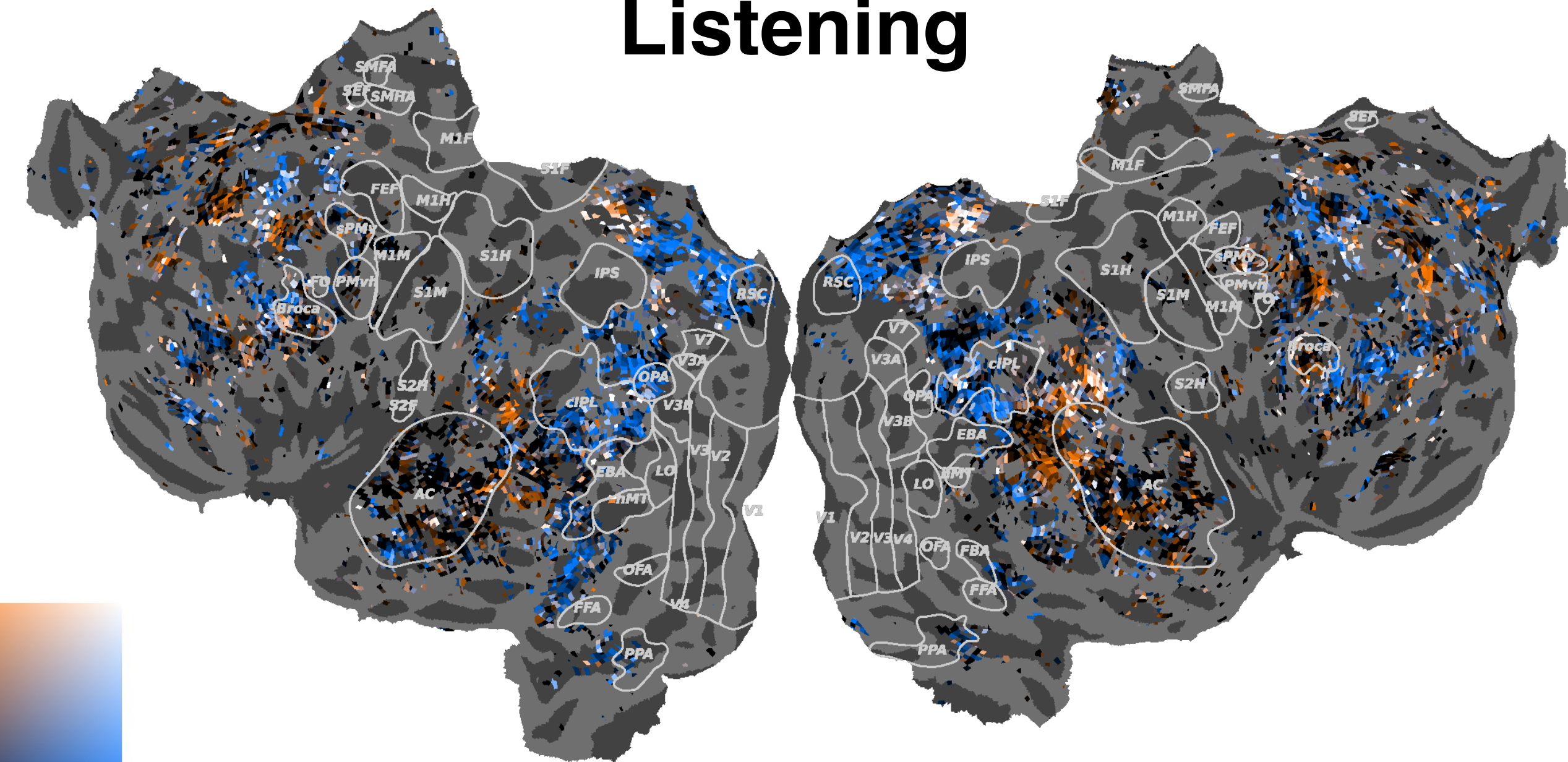


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Reading



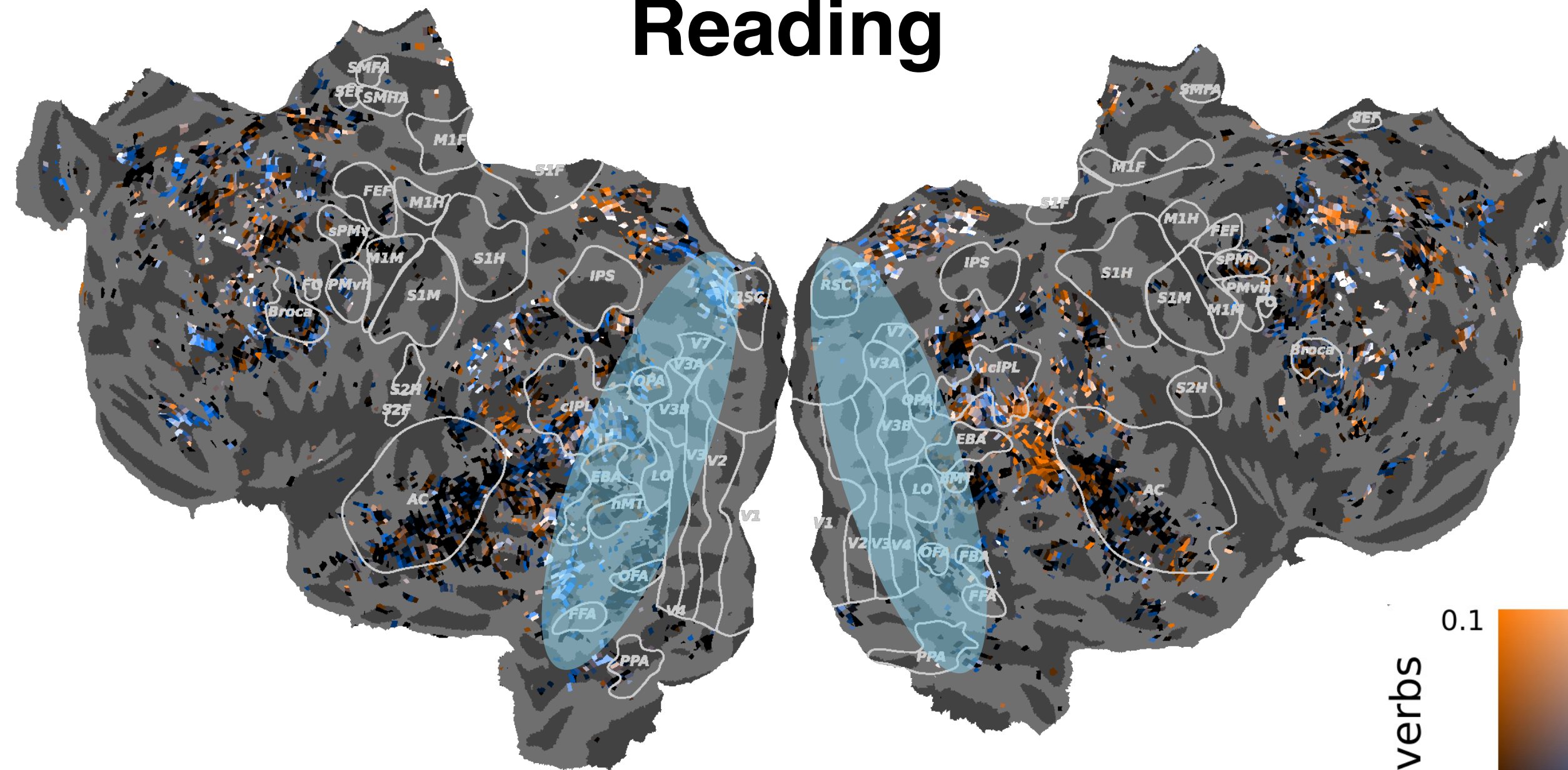
Subject 7  
Listening



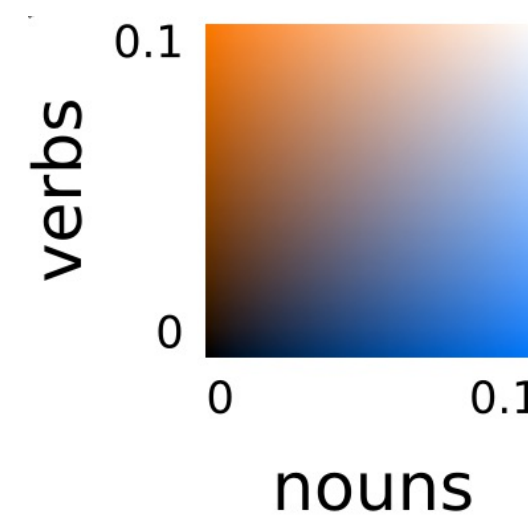
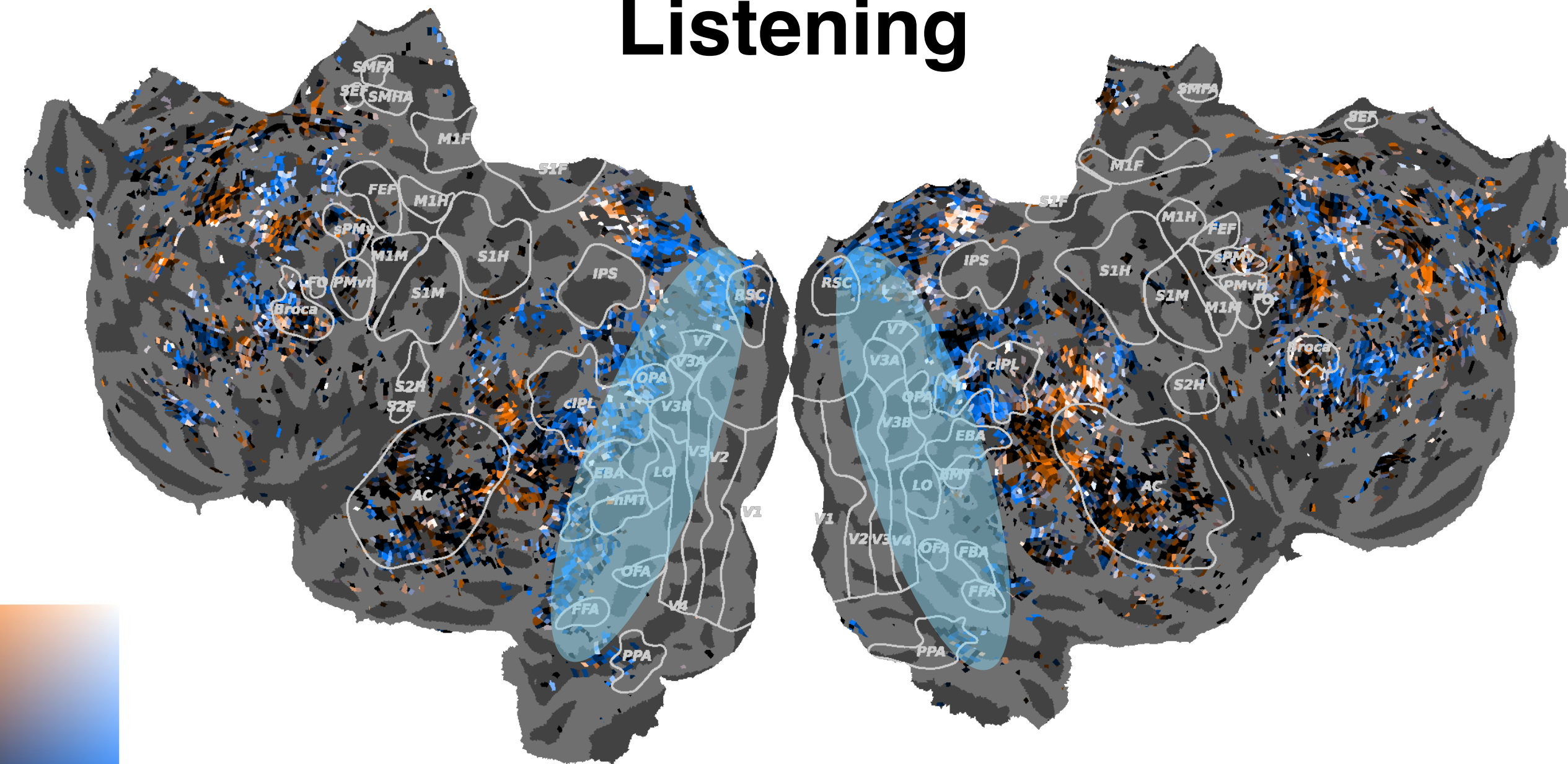


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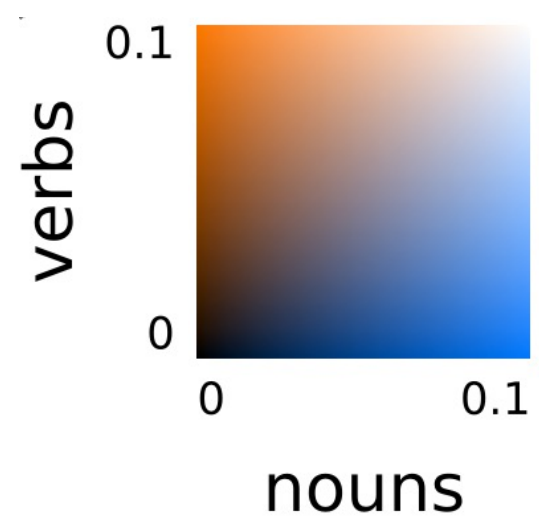




Technische  
Universität  
Berlin

# Subject 7

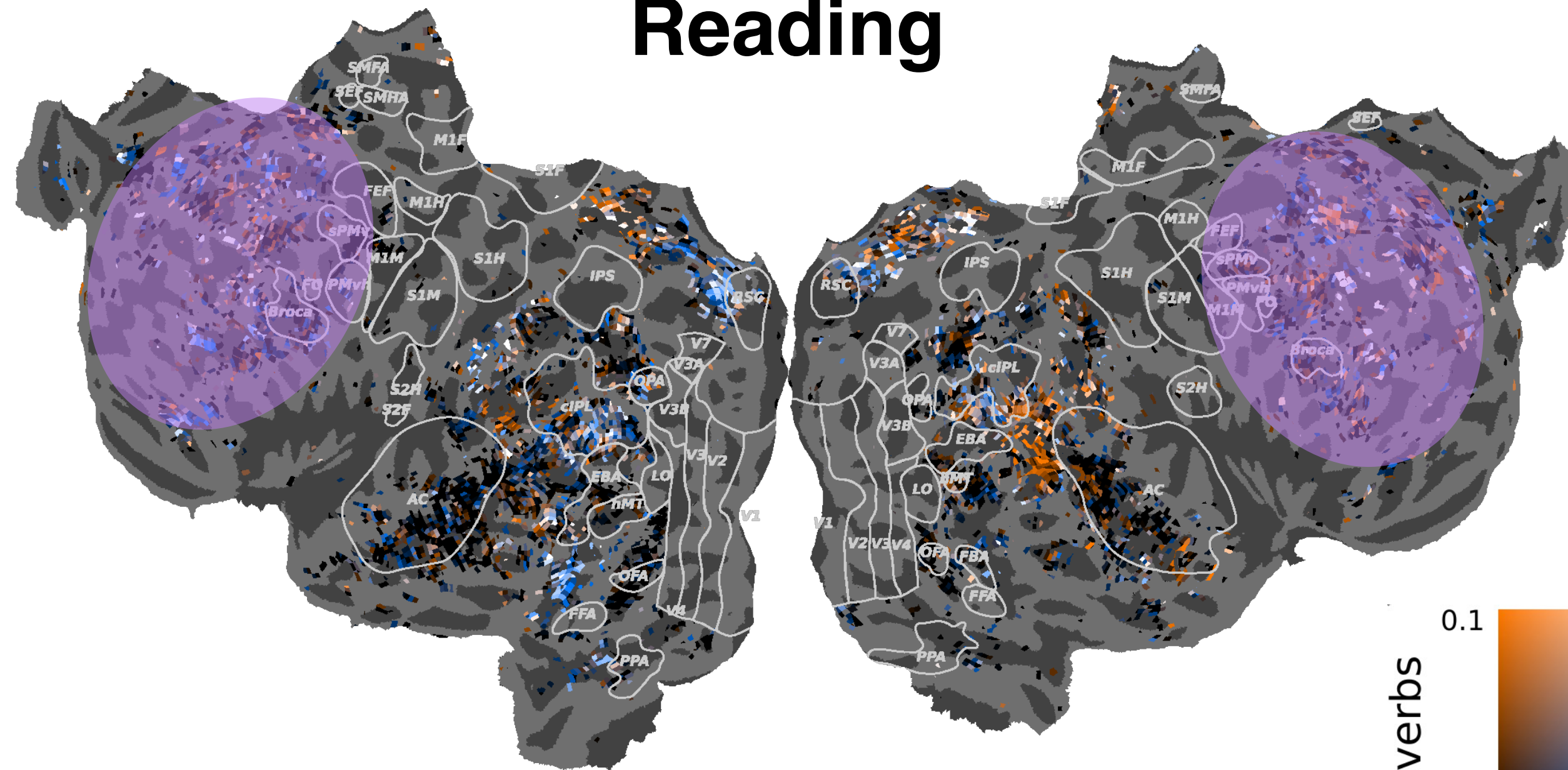
## Listening



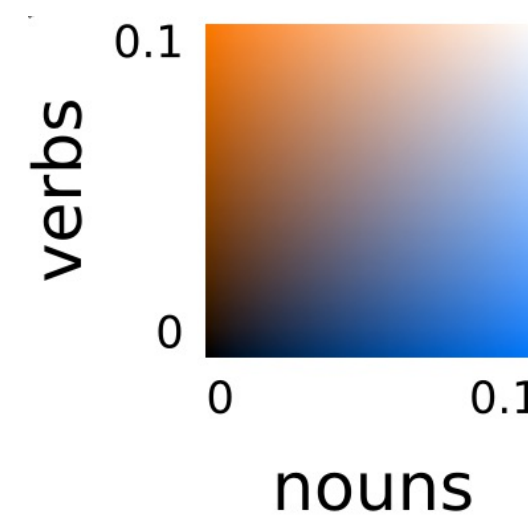
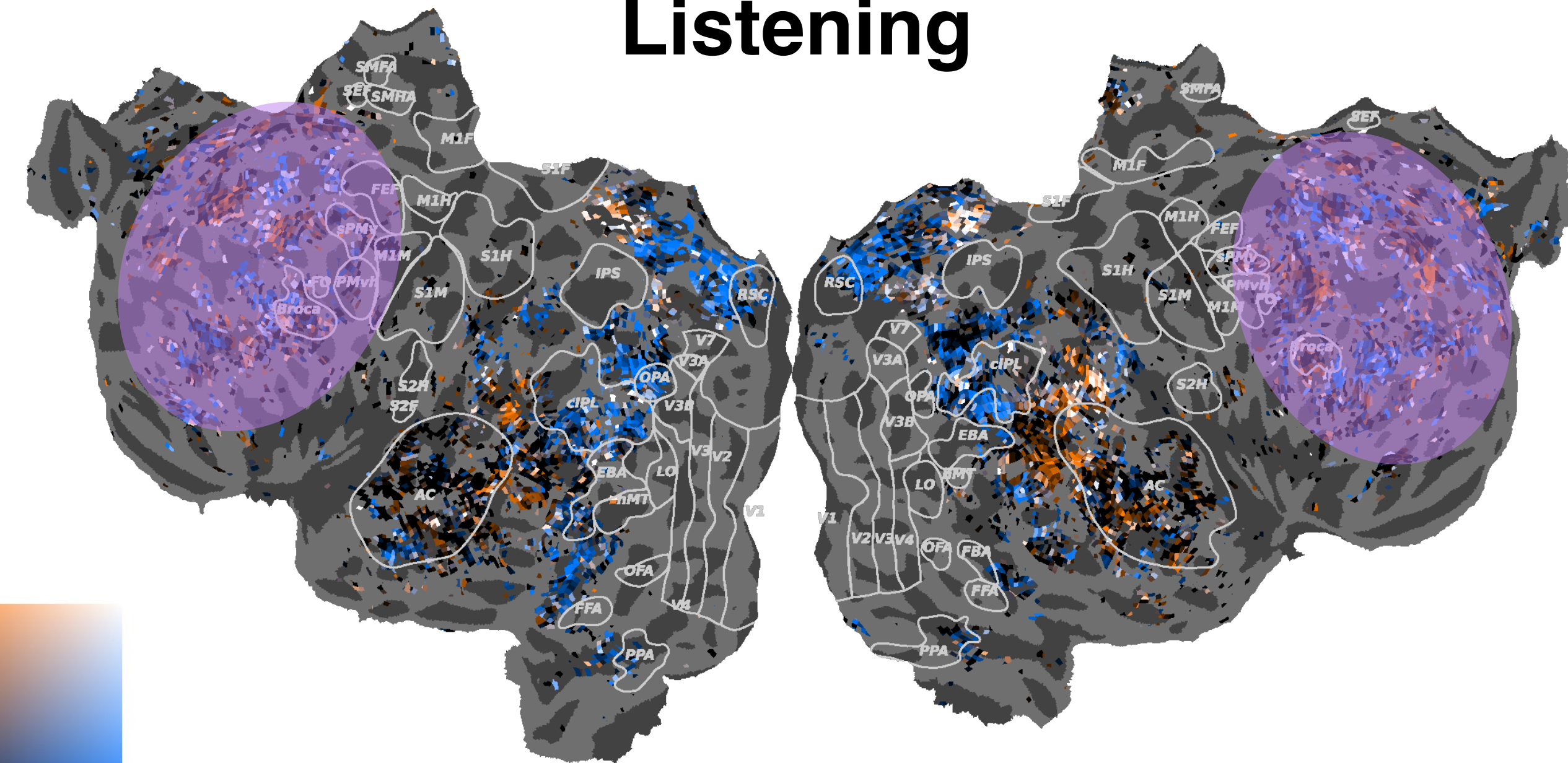


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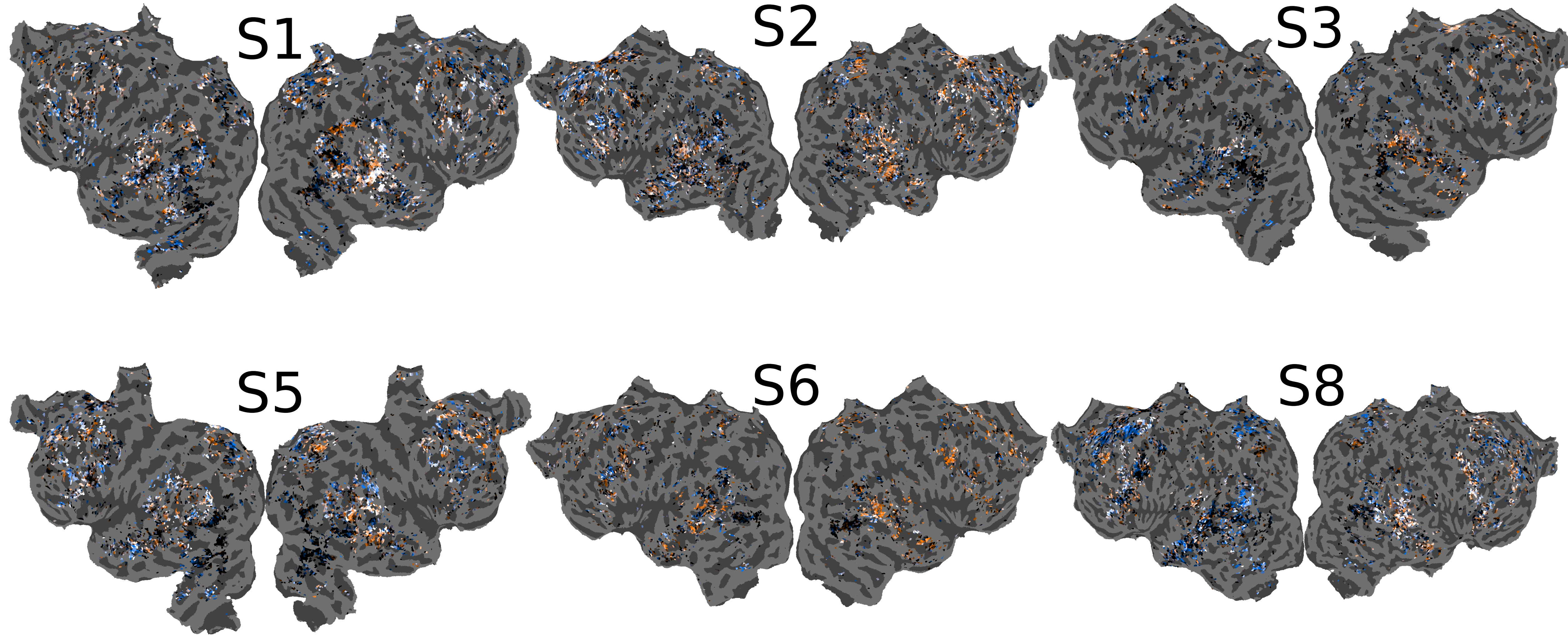


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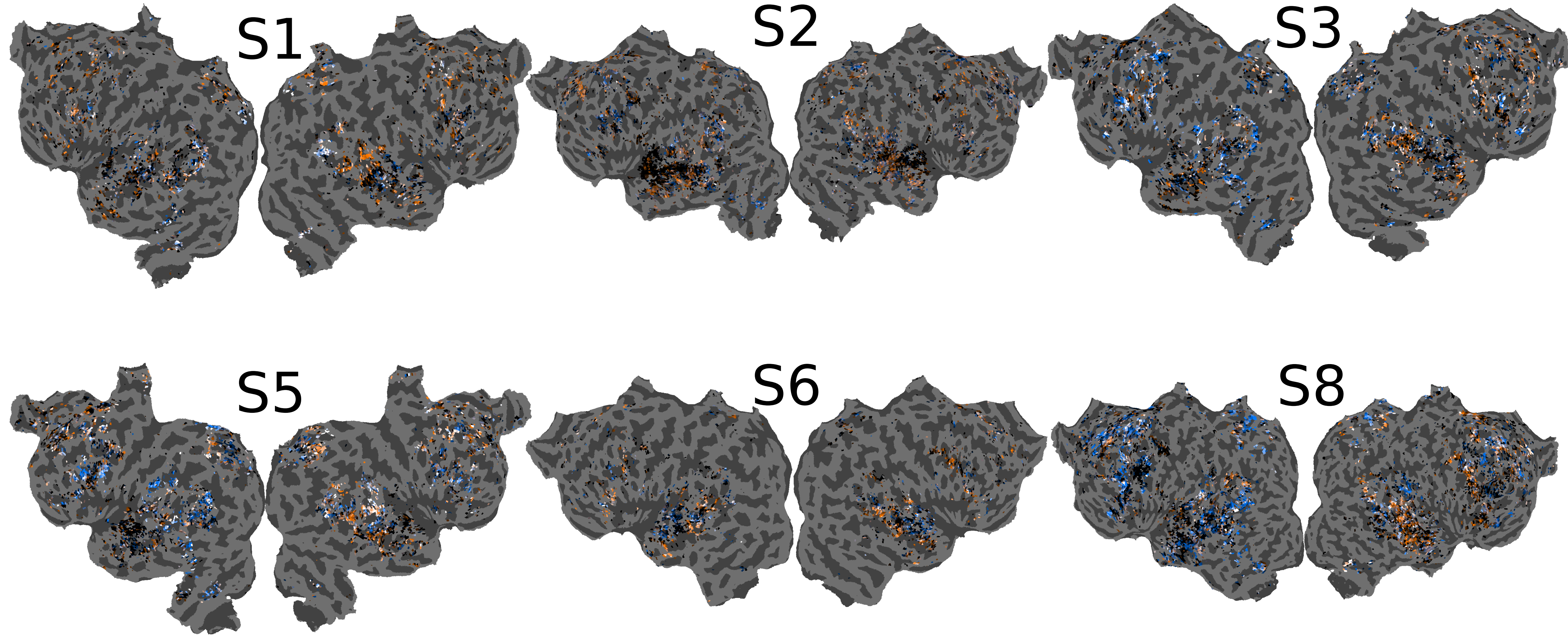


# NOUN AND VERB SENSITIVITY – READING





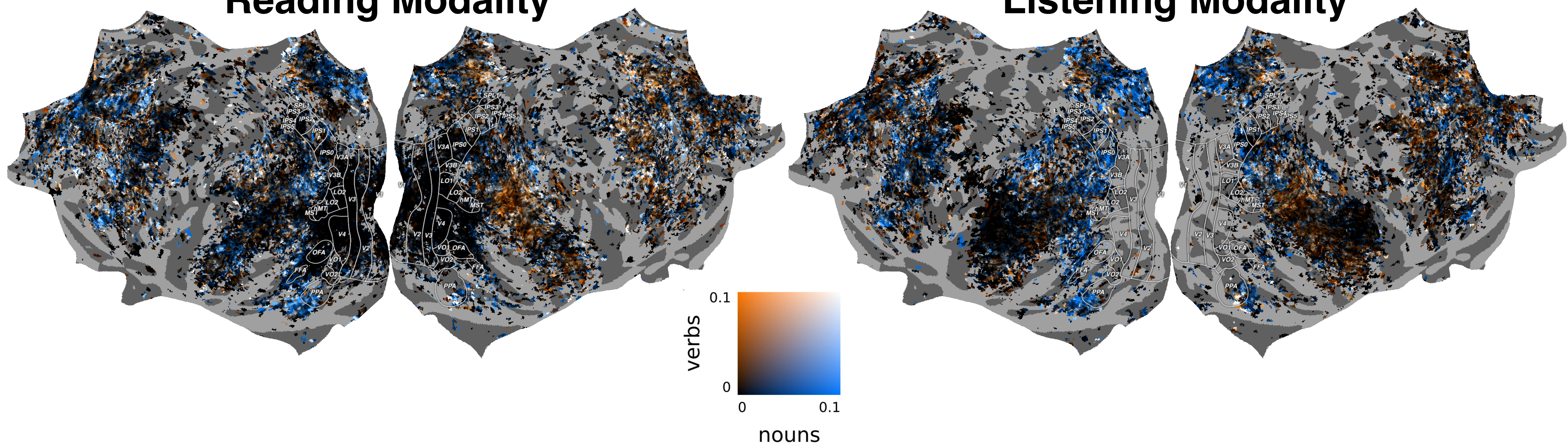
# NOUN AND VERB SENSITIVITY – LISTENING





Reading Modality

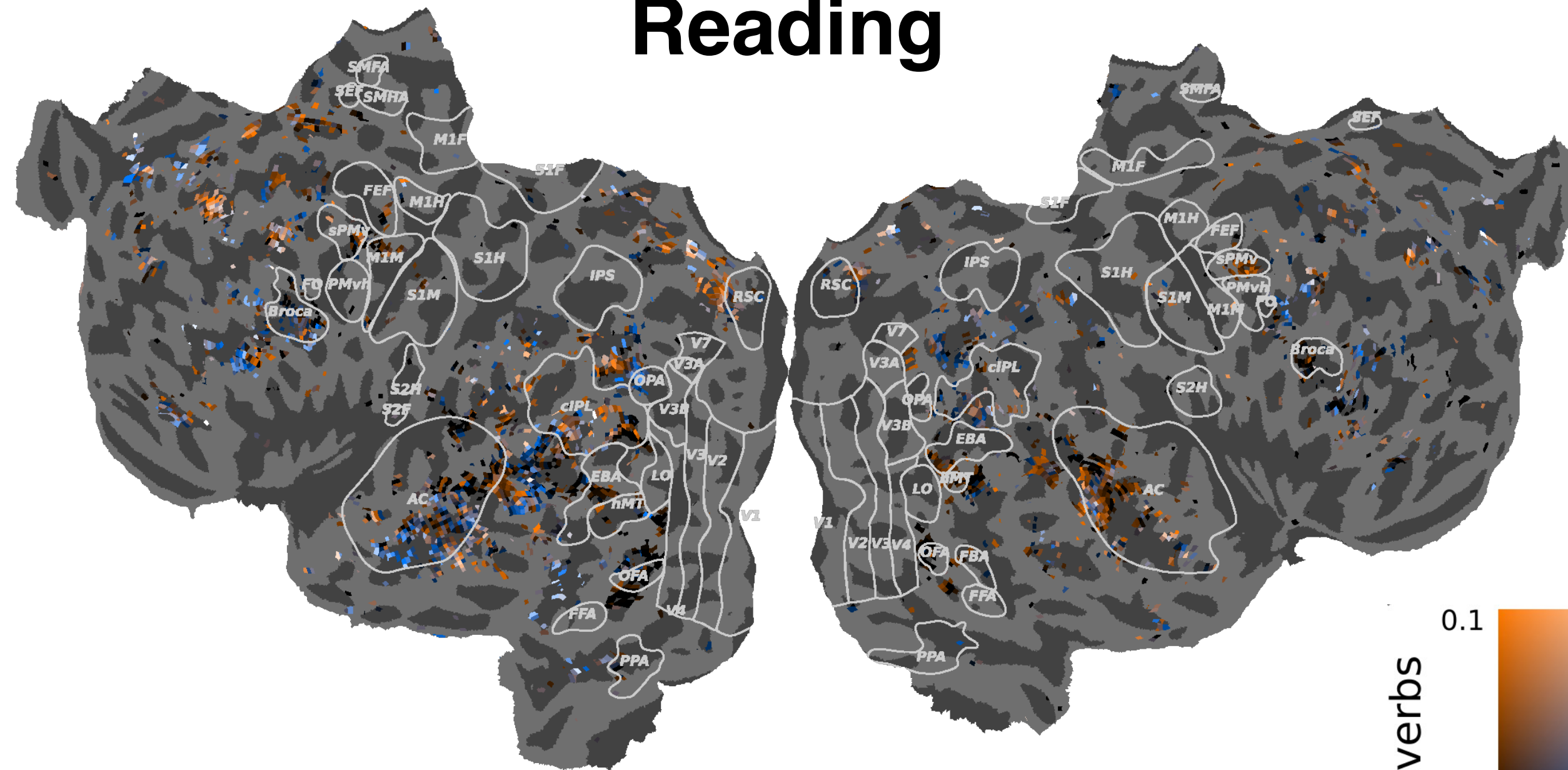
Listening Modality



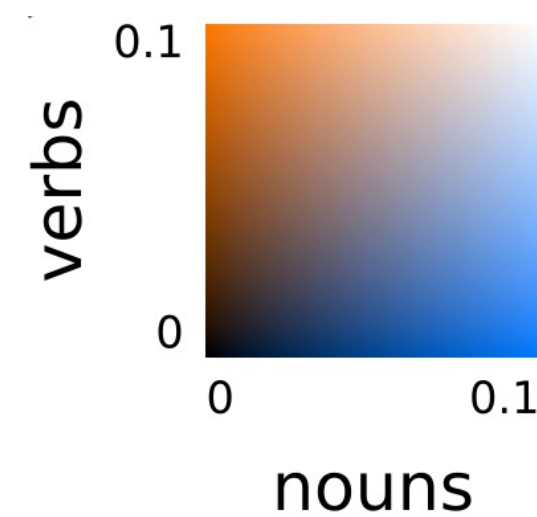
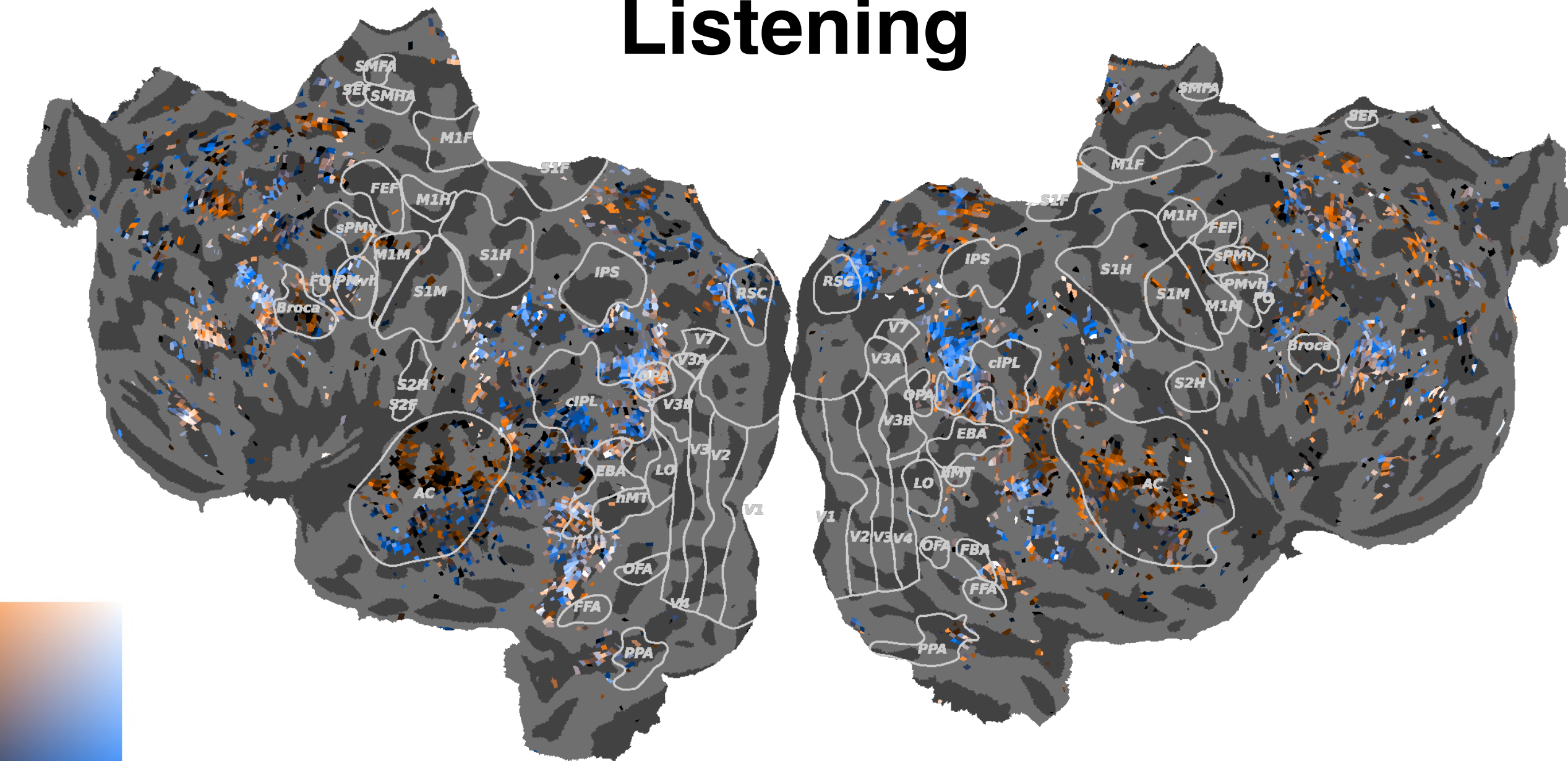


# MASKING OF LEXICAL EMBEDDINGS PRODUCES SIMILAR SENSITIVITY MAPS

Subject 7  
Reading



Subject 7  
Listening

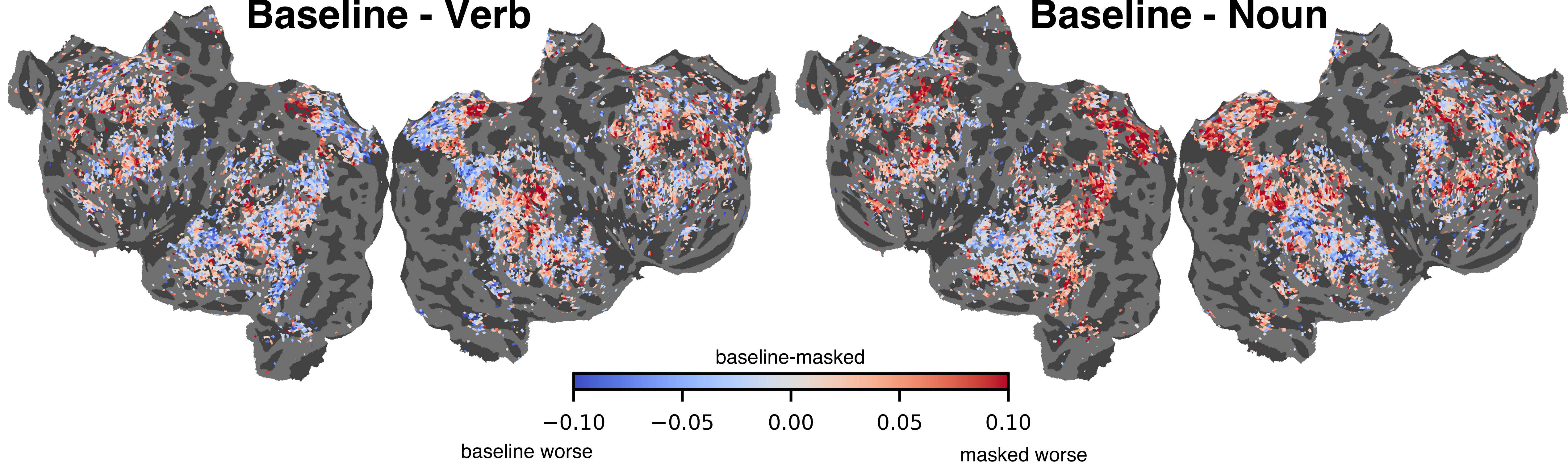




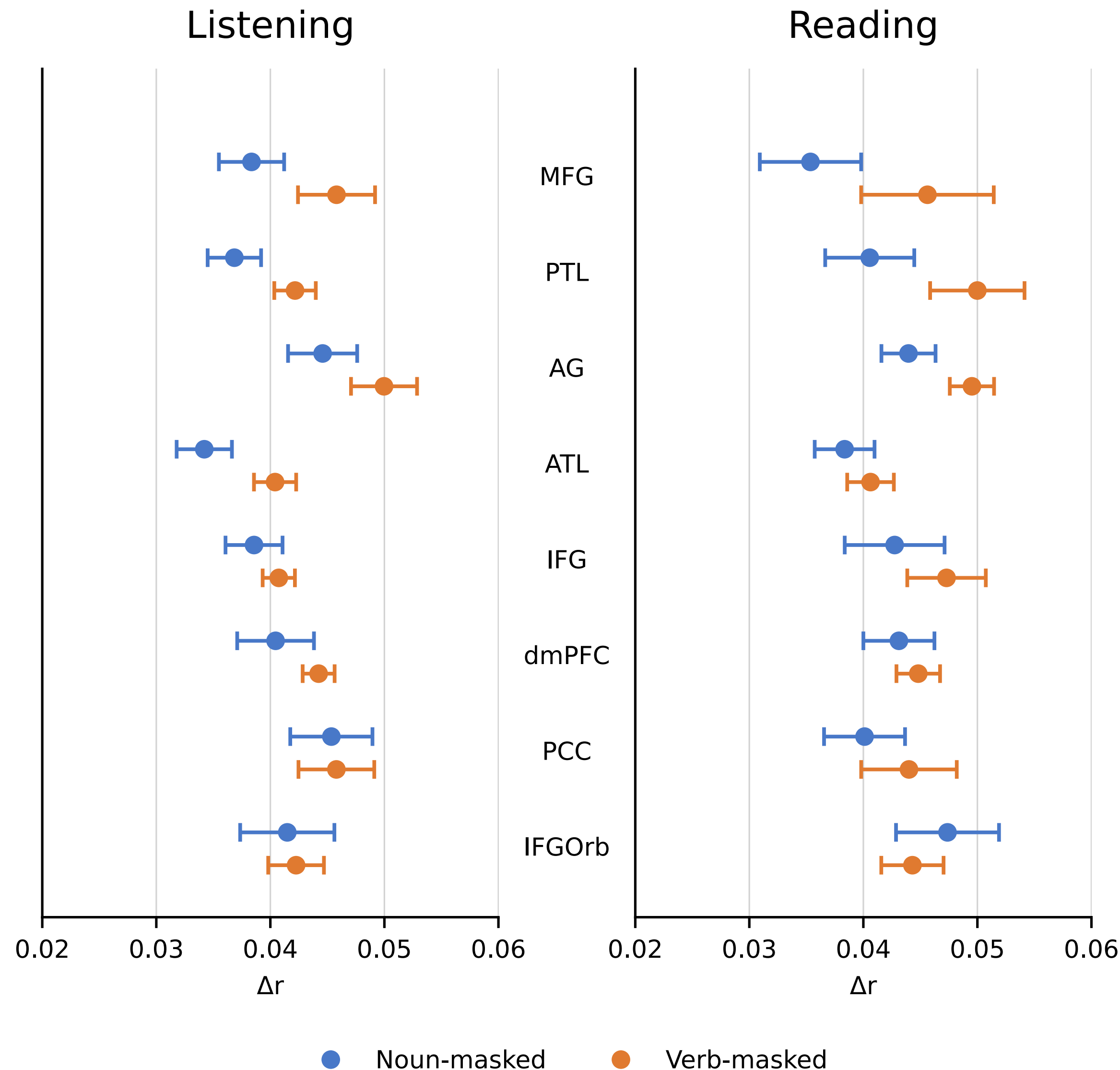
# MASKING NOUNS AND VERBS IMPROVES PREDICTION ACCURACY IN SOME AREAS

**Subject 7**  
**Baseline - Verb**

**Subject 7**  
**Baseline - Noun**



# VERB SENSITIVITY IS MORE PREVALENT ACROSS LANGUAGE ROIS



## Language ROIs

Fedorenko et al. (2010)

## Mean Drop

0.03 – 0.06

## Verb > Noun

MFG, PTL, AG, ATL

## Reading vs Listening

Similar across ROIs



# COMPARISON WITH LITERATURE

| Theoretical account                                                                                        | Predicted pattern                                                                  | Observed pattern                                                       |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Fronto-Temporal View</b><br><i>Neuropsychological</i><br>(A. R. Damasio and Tranel 1993)                | V > N in frontal regions;<br>N > V in temporal regions                             | V > N in temporal regions (ATL, PTL); no clean frontal/temporal split. |
| <b>Emergentist View</b><br><i>Object-Action, Imageability</i><br>(Bird et al. 2000; Vigliocco et al. 2011) | V > N in frontal and parietal regions; N > V in inferior temporal regions.         | N > V near visual areas; Similar sensitivity in frontal regions.       |
| <b>Combinatorial</b><br><i>Morphology/inflection</i><br>(Tyler et al. 2004; K. A. Shapiro et al. 2005)     | N = V broadly, with V > N specifically in IFG for morphological composition.       | Slight V > N in IFG; N > V in IFGOrb.                                  |
| <b>Distributed</b><br><i>Shared network</i><br>(Crepaldi et al. 2013; Caucheteux et al. 2021)              | Noun and verb circuits distributed across frontal, parietal, and temporal regions. | Smaller clusters of noun and verb sensitive voxels across the brain.   |

# CONCLUSIONS

Noun and verb sensitivity forms **distinct** but **distributed clusters** across frontal, parietal and temporal cortex



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Verb sensitivity dominates across most language ROIs

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Results support a **distributed account**



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Only the **absence** of nouns / verbs is being investigated

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Cross-Modality **not confirmed**

# FUTURE RESEARCH

Investigating **presence** of nouns and verbs using one-hot encodings



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Using a **different dataset** other than Moth



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Looking at a **different Language** other than English, e.g. Chinese

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Cluster categorization: What **kind** of nouns / verbs in the clusters?